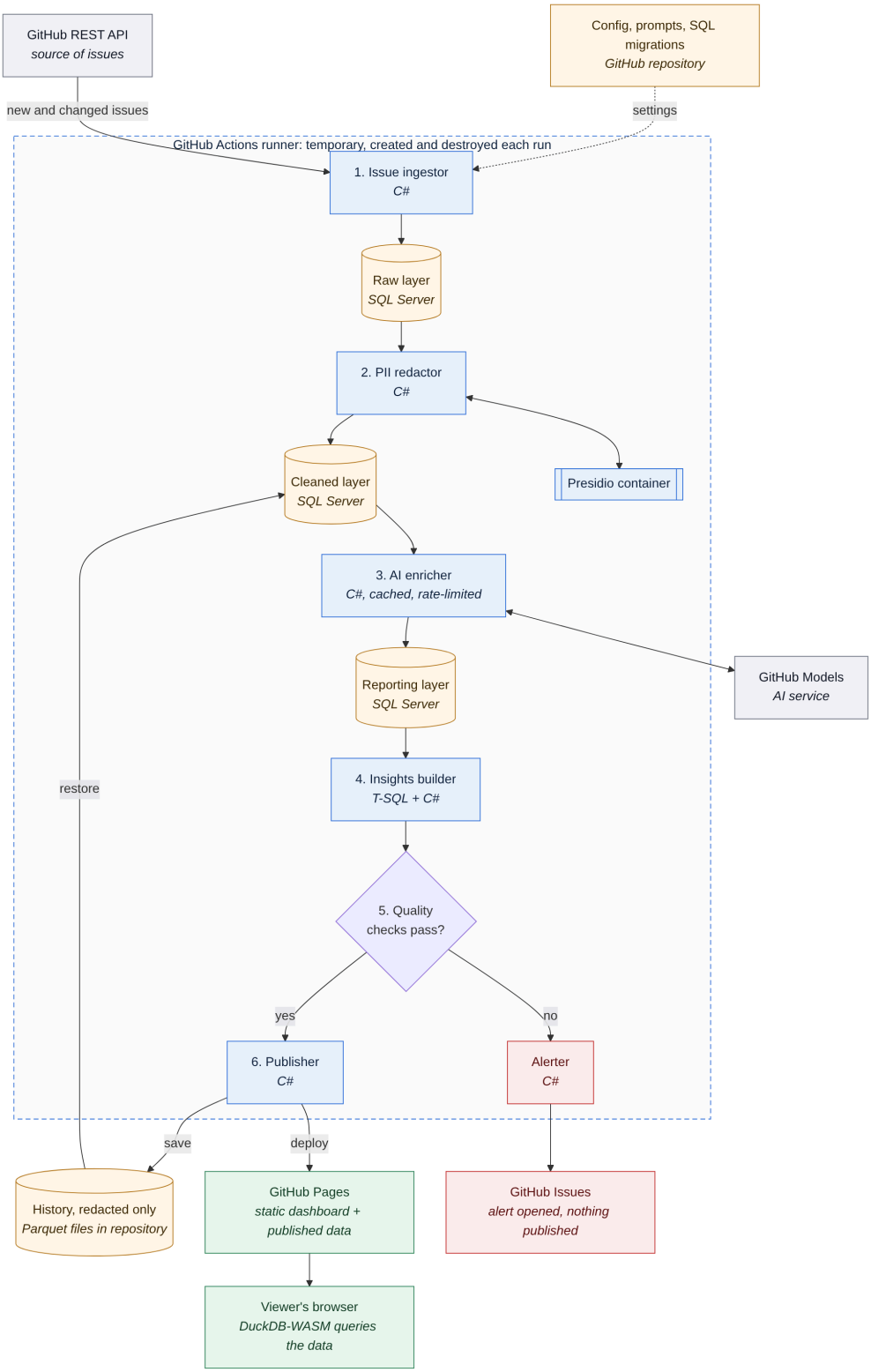
# Design: Feedback Intelligence Platform

| | |
|---|---|
| **Status** | Approved |
| **Version** | 1.0 |
| **Date** | 2026-10-18 |
| **Owner** | Benjamin Rose |
| **Related** | Project plan, Requirements, Data dictionary, Decision records |

This document describes how the Feedback Intelligence Platform is built: its architecture, components, failure handling, configuration, dashboard, security, and repository layout. Detailed table definitions are in the data dictionary, and the reasoning behind each major choice is in the decision records.

## 1. Architecture

A scheduled GitHub Actions workflow runs a C# pipeline once a day. Each run starts temporary SQL Server and Presidio containers, restores history, processes new and changed issues through raw, cleaned, and reporting layers, and publishes a static dashboard to GitHub Pages. The runner is destroyed afterward, so nothing survives except what is deliberately saved.




*Source:* `diagrams/architecture.mmd` *(Mermaid).*

| Environment | What runs there |
|---|---|
| **GitHub Actions runner (temporary)** | The C# pipeline, SQL Server Express, and Presidio containers |
| **GitHub repository** | Code, config, prompts, migrations, docs; history on the `data` branch |
| **GitHub Pages** | The static dashboard and its published data files |
| **Viewer's browser** | Dashboard code and DuckDB-WASM |
| **External services** | GitHub REST API (issues) and GitHub Models (AI) |

**Key decisions:** ADR 0002 (GitHub as platform), 0004 (temporary SQL Server), 0005 and 0013 (history), 0009 (data layers).

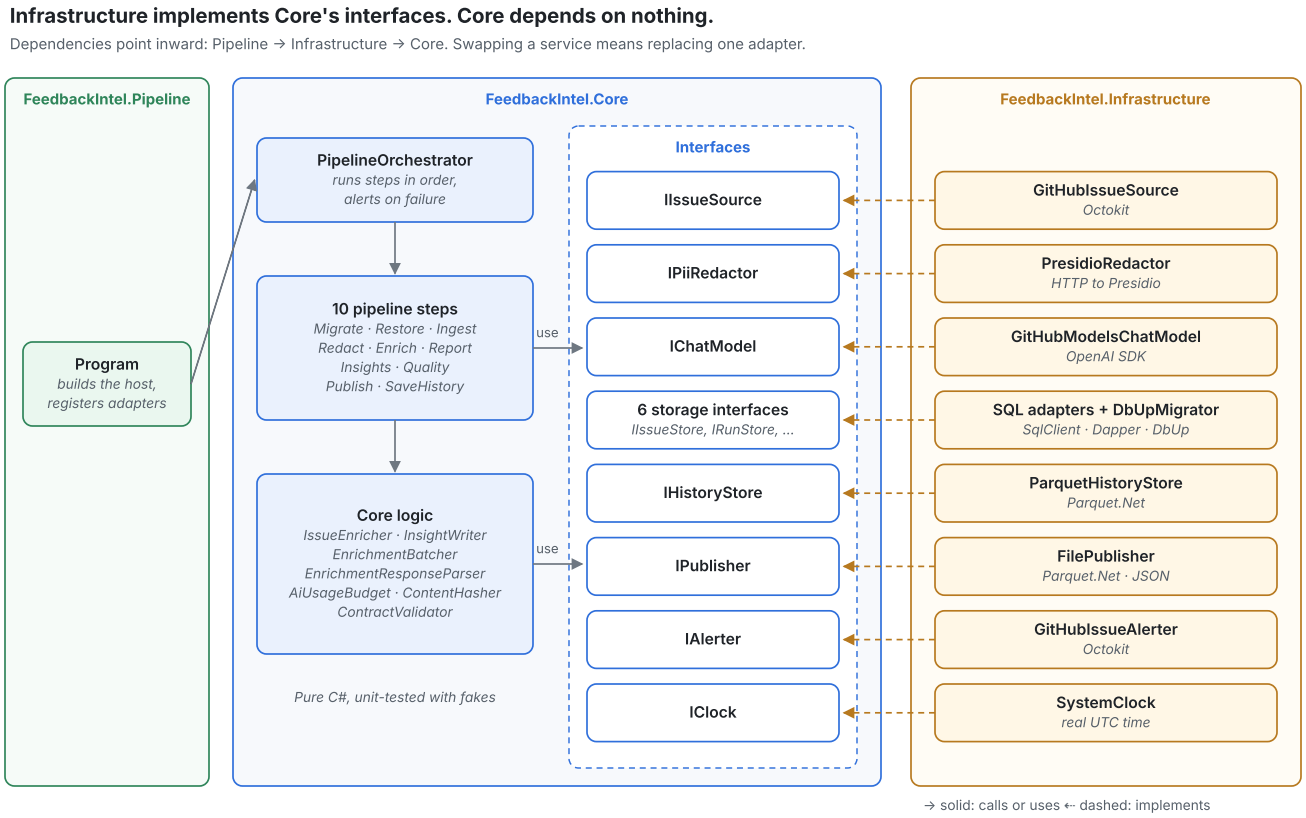
## 2. Data design

Data moves through four SQL Server schemas:

| Schema | Holds | Saved as history | Published |
|---|---|---|---|
| `raw` | Issues as fetched, unredacted | Never | Never |
| `cleaned` | Redacted issues, AI results, digests | Yes (without issue bodies) | No |
| `reporting` | Aggregates and dashboard tables, rebuilt each run | No | Yes |
| `ops` | Run records and watermarks | Yes | Run health only |

The **data contract** lists every field allowed in published and history files; an automated test enforces it. Full definitions: data dictionary.

# 3. Components




| Project | Responsibility |
|---|---|
| `FeedbackIntel.Core` | Data models, interfaces, pipeline steps, and pure logic (batching, validation, hashing, data contract). No external dependencies, so it is fully unit-testable with fakes. |
| `FeedbackIntel.Infrastructure` | Adapters that implement Core's interfaces using real services: Octokit, Presidio over HTTP, the OpenAI SDK for GitHub Models, SqlClient with Dapper and DbUp, and Parquet.Net. |
| `FeedbackIntel.Pipeline` | Console entry point. Loads configuration, registers adapters through dependency injection, and starts the orchestrator. |
| `FeedbackIntel.Core.Tests` | Fast unit tests using fakes. |
| `FeedbackIntel.Integration.Tests` | Tests against real SQL Server and Presidio containers via Testcontainers. |
| `FeedbackIntel.Evaluation` | Scores prompts against the labeled evaluation set (FR-DEV-02). |

**Dependency rule:** Infrastructure depends on Core; Core depends on nothing. Every external service sits behind an interface, so it can be faked in tests or replaced by another provider (NFR-MNT-03). Class-level documentation lives in the code as XML doc comments.

**Interfaces:** `IPipelineStep`, `IIssueSource`, `IPiiRedactor`, `IChatModel`, `IIssueEnricher`, `IInsightWriter`, `IDatabaseMigrator`, `IIssueStore`, `IEnrichmentStore`, `IRunStore`, `IReportingBuilder`, `IQualityChecker`, `IHistoryStore`, `IPublisher`, `IAlerter`, `IClock`.

## 4. Pipeline flow and error handling

### Rules

1. **Temporary errors are retried; permanent errors are not.** Network errors, timeouts, and rate-limit responses are retried up to 3 times with increasing waits (2, 4, 8 seconds). Bad data, bad configuration, and code errors fail immediately.
2. **Any failure stops the run.** The orchestrator records the failed step, saves the run record, raises an alert, and exits with a failure code. Nothing later runs, so nothing half-finished is published.
3. **A failed run is never harmful.** The dashboard keeps the last good data, and the watermark doesn't move, so the next run picks up everything missed.
4. **Timeouts:** the pipeline step has a 40-minute limit inside a 50-minute job, leaving time for the failure-alert step. The normal run target is 30 minutes (NFR-PERF-03).
5. **Logging:** every log entry carries the run ID and step name automatically (NFR-OBS-01). Logs never include issue text, because run logs of a public repository are public.

### Steps

| # | Step | On failure | Retried? |
|---|---|---|---|
| — | Freshness check (after restore) | Stale-data alert if data is older than 48 hours; the run continues | — |
| 1 | Migrate database | Alert | Waits up to 2 minutes for SQL Server; script errors not retried |
| 2 | Restore history | Alert, unless it is a manual **bootstrap** run | No |
| 3 | Ingest issues | Alert; also fails if more than 10% of records are rejected | Yes; waits for a rate-limit reset only if under 15 minutes |
| 4 | Redact | Alert; **never continues without redaction** | Yes |
| 5 | Enrich | Not a run failure: invalid answers become `unclassified`; unprocessed issues stay `pending` | Yes; requests spaced about 4 seconds apart; every attempt counts against the daily budget |
| 6 | Rebuild reporting | Alert | No |
| 7 | Build insights | Not a run failure: topics show "unavailable today"; a missing digest is generated on the next run | Yes |

| # | Step | On failure | Retried? |
|---|---|---|---|
| 8 | Quality checks | Alert listing every failed check; nothing published | No |
| 9 | Publish (write files) | Alert; contract violations are treated as possible privacy issues | Write errors yes; violations no |
| 10 | Save history (write files) | Alert | Yes |
| — | Workflow: deploy Pages, push history | Workflow-level alert, if the pipeline hasn't already raised one | Yes |

**Two deliberate exceptions:** AI problems never fail the run, because the AI is an enhancement and the dashboard should still update. Redaction problems always fail it, because privacy is never traded for availability.

## 5. Configuration

All tunable behavior lives in `config/pipeline.json`, validated against `config/pipeline.schema.json` in CI and at startup (FR-CFG-01, FR-CFG-02). Any value can be overridden by an environment variable, which is how local runs redirect output away from the live site.

```
{
  "Repositories": [
    { "Owner": "microsoft", "Name": "winget-cli", "Enabled": true },
    { "Owner": "microsoft", "Name": "PowerToys",  "Enabled": false },
    { "Owner": "microsoft", "Name": "vscode",     "Enabled": false }
  ],
  "Ingestion":      { "InitialHistoryDays": 90, "MaxRejectedPercent": 10,
"RateLimitMaxWaitMinutes": 15 },
  "Classification": { "Categories": [ "bug", "feature_request", "question", "docs",
"performance" ],
                      "Severities": [ "low", "medium", "high", "critical" ] },
  "Ai":             { "Endpoint": "https://models.github.ai/inference", "Model":
"openai/gpt-4.1-mini",
                      "PromptDirectory": "prompts/v1", "DailyRequestLimit": 140,
                      "MinSecondsBetweenRequests": 4, "MaxIssuesPerRequest": 10,
                      "MaxCharactersPerRequest": 24000, "MaxRetries": 3 },
  "Text":           { "BodyMaxCharacters": 4000, "SummaryMinBodyCharacters": 500 },
  "Insights":       { "TopicMinIssues": 3, "TopicWindowDays": 7, "DigestMaxWords": 300 },
  "Operations":     { "StaleDataAlertHours": 48, "SqlStartupWaitSeconds": 120 },
  "Publishing":     { "OutputDirectory": "out/site", "HistoryDirectory": "out/history" }
}
```

| Not in this file | Where it lives | Why |
|---|---|---|
| **Schedule and timeouts** | Workflow file | GitHub reads these only from workflow files |
| **Tokens and passwords** | Environment variables | Secrets never enter the repository (NFR-SEC-01) |
| **Database connection string** | Environment variable | Differs between local, CI, and production |

**Notes:**

- **Categories and severities come from config, not code enums,** so they can change through a pull request (US-12). Because changing them also affects the prompts and evaluation labels, CI checks that all three agree.
- **The model name is an example,** to be confirmed in M4 against the evaluation set.
- **The daily AI limit is 140,** deliberately below GitHub's limit of about 150 to leave a safety margin.

## 6. Dashboard

A static site of plain HTML and JavaScript, with charts drawn by Chart.js and queries run in the browser by DuckDB-WASM (ADR 0008).

| Page | Shows | Reads | Milestone |
|---|---|---|---|
| **Overview** | Last 7 days vs. previous 7: new, closed, % bugs, critical, median days to close | `site_info.json`, `overview.json` | M3 |
| **Trends** | Weekly volume, closures, and category mix; repository and date filters | `weekly_metrics.parquet`, `weekly_category.parquet` | M3 |
| **Issue explorer** | Searchable, filterable issue table with links to GitHub; CSV export (M8) | `issues.parquet` | M3 |
| **Emerging topics** | Ranked fast-growing topics; select one to list its issues | `topics.parquet`, `topic_issues.parquet`, `issues.parquet` | M5 |
| **Weekly digest** | Latest digest and archive | `digests.parquet` | M5 |
| **Pipeline health** | Last 30 runs, data freshness, AI usage vs. limit | `run_health.parquet`, `site_info.json` | M5 |

**Loading:** the Overview page renders immediately from two small JSON files; DuckDB-WASM loads in the background and is needed only for the other pages. This addresses the 3-second load target (NFR-PERF-02).

**Every page shows:** a repository selector, the time the data was last updated (FR-DSH-05), and an "AI-generated" marker next to AI categories, severities, summaries, topics, and digests (FR-DSH-06).

**Accessibility:** responsive layout for desktop and mobile (NFR-USE-01); WCAG AA contrast, and charts that never rely on color alone (NFR-USE-02).

## 7. Security and privacy

| Threat | Mitigation |
|---|---|
| **Personal information in issue text reaches the AI, history, or dashboard** | Presidio redaction before anything leaves the raw layer; usernames never collected; bodies never saved or published; data contract test; PII test set in CI (FR-PRV-01 to 04, ADR 0013) |
| **Issue text printed in public run logs** | Logs record counts, IDs, and step names only, never issue text. Run logs of a public repository are visible to anyone. |
| **Malicious HTML or script in issue titles or AI output** (cross-site scripting) | The dashboard inserts all data as plain text, never as HTML, and the site sets a Content Security Policy that blocks inline scripts. |
| **Prompt injection through issue text** | Issue text is clearly marked as data in prompts; AI output is parsed as JSON and validated against configured values; the AI has no ability to act (NFR-SEC-03, risk R4) |
| **Leaked credentials** | Only the built-in, short-lived `GITHUB_TOKEN` is used; secret scanning in CI (NFR-SEC-01) |
| **Over-privileged workflow** | Each job declares minimum permissions; only the alert step can write issues, and only the deploy step can write Pages (NFR-SEC-02) |
| **Compromised dependency or third-party action** | Dependency vulnerability scanning in CI; third-party GitHub Actions pinned to exact commit versions (NFR-SEC-04) |

| Threat | Mitigation |
|---|---|
| **Incorrect or harmful data published** | Quality checks gate publishing; the contract validator blocks unexpected fields; failures publish nothing |

## 8. Repository structure

```
feedback-intelligence/
├── README.md                    ← front door: what it is, live link, how to run
├── CONTRIBUTING.md              ← workflow, standards, AI-assisted development guide
├── .gitattributes               ← consistent line endings (ADR 0012)
├── .github/
│   ├── workflows/               ← ci.yml, pipeline.yml, evaluate.yml
│   └── pull_request_template.md ← self-review checklist and AI disclosure
├── config/                      ← pipeline.json and its schema
├── prompts/v1/                  ← versioned AI prompts
├── database/migrations/         ← numbered .sql scripts (DbUp)
├── src/
│   ├── FeedbackIntel.Core/
│   ├── FeedbackIntel.Infrastructure/
│   └── FeedbackIntel.Pipeline/
├── tests/
│   ├── FeedbackIntel.Core.Tests/
│   ├── FeedbackIntel.Integration.Tests/
│   └── FeedbackIntel.Evaluation/   ← includes the labeled evaluation set
├── dashboard/                   ← static site source
├── deploy/                      ← Bicep templates, validated in CI only (M7)
├── docker-compose.yml           ← local SQL Server and Presidio
└── docs/                        ← plan, requirements, design, data dictionary, ADRs,
diagrams
```

The `data` branch holds only history files and is written only by the pipeline workflow.

## 9. Open items to verify during the build

| Item | When |
|---|---|
| **Whether daily commits to the `data` branch keep scheduled workflows active (ADR 0005, risk R6)** | M2 |
| **SQL Server container startup time on GitHub runners (assumption A4)** | M2 |
| **Presidio accuracy on real issues, including custom recognizers (assumption A5)** | M3 |
| **Dashboard load time with DuckDB-WASM (NFR-PERF-02)** | M3 |
| **Model choice and prompt accuracy against the evaluation set (assumption A6)** | M4 |

## Revision history

| Version | Date | Change |
|---|---|---|
| **1.0** | 2026-10-18 | Initial design approved |